Session 02 - Finishing the Caesar Cipher - Tests

Python Contoso Vault - Session 02

Part 1 - Core: shift.py

Goal

Read a message and a key. Encrypt the message, then decrypt it again to prove the round trip works. Keep the case of each letter. Leave every character that is not a letter unchanged.

Run `python shift.py` in the terminal. Type each input line and press Enter after it.

Instructions

The class builds steps 1 to 10 for capital letters only, then commits the file. Finish at home: make steps 3, 4, 7 and 8 work for lowercase letters too.

1. Read a message as a string, with the prompt `"Message: "` on the same line.
2. Read a key from 0 to 25 as an integer, with the prompt `"Key: "` on the same line.
3. Shift each letter of the message forward by the key, within its own case.
4. Wrap any shifted letter that passes the end of its alphabet.
5. Leave every character that is not a letter unchanged.
6. Print the ciphertext as `Ciphertext: <text>`, with one space after the colon.
7. Shift each letter of the ciphertext back by the key.
8. Wrap any shifted letter that passes the start of its alphabet.
9. Print the plaintext as `Plaintext: <text>`, with one space after the colon.
10. Prove that the plaintext matches the message.

Constraints

- Read each value with a prompt on the same line, as in `input("Message: ")`. The prompts print in this order: `"Message: "` for the message, then `"Key: "` for the key. Each prompt is its label, a colon and one trailing space. Type each prompt exactly as shown. A program with different prompt text does not meet the test plan.
- After the two prompts, the output is exactly two lines: `Ciphertext: <text>`, then `Plaintext: <text>`. Put one space after each colon.

Design

The following rules say how to write the code. Follow them.

- Wrap shifted letters with an `if`, without `%`.
- Prove the round trip with an `assert`.
- Give every variable a type hint.

Input validation

Not required. Assume the key is always a whole number from 0 to 25. Any message is valid, including an empty one.

Test plan

The program prints both prompts first, so the expected output in the following test plan shows the two lines printed after the prompts.

Test	Message	Key	Expected output	Actual output	Pass
1	`Meet at Contoso HQ!`	`3`	`Ciphertext: Phhw dw Frqwrvr KT!` `Plaintext: Meet at Contoso HQ!`		
2	`Zebra zone 42, room 7.`	`25`	`Ciphertext: Ydaqz ynmd 42, qnnl 7.` `Plaintext: Zebra zone 42, room 7.`		
3	`Contoso HQ`	`0`	`Ciphertext: Contoso HQ` `Plaintext: Contoso HQ`		
4	`Contoso HQ`	`1`	`Ciphertext: Dpouptp IR` `Plaintext: Contoso HQ`		
5	`Contoso HQ`	`13`	`Ciphertext: Pbagbfb UD` `Plaintext: Contoso HQ`		
6	`Contoso HQ`	`25`	`Ciphertext: Bnmsnrn GP` `Plaintext: Contoso HQ`		
7	`2026-11-01, 9:00!`	`7`	`Ciphertext: 2026-11-01, 9:00!` `Plaintext: 2026-11-01, 9:00!`		
8	`mIxEd CaSe`	`4`	`Ciphertext: qMbIh GeWi` `Plaintext: mIxEd CaSe`		
9	empty line	`5`	`Ciphertext:` `Plaintext:`		

Done criteria

- ☐ Every test plan row prints exactly the expected lines.
- ☐ The script runs from the terminal with `python shift.py`.

Marking

This take-home is marked out of 100, with a pass mark of 60.

Part	Tests	Design	Total
Core	80	20	100
Total	**80**	**20**	**100**

Guidelines

- Test marks show that the program works. The instructor's tests run it on every row of the test plan, plus further cases.
- Design marks show that the code is built well. The instructor reads the code against each Design rule.
- Marking uses the last commit pushed by 11:59 PM on the day before the next session.
- The submission is one GitHub link to the fork, posted on the portal.